



The Müntz spectral element method for the numerical solutions of regularized long wave equation

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ABSTRACT

This paper introduces new bases derived from Müntz polynomials for application in the spectral element method. Within this framework, we demonstrate that the mass and stiffness matrices exhibit matrix with three non-zero diagonals and diagonal structures, respectively. The Müntz spectral element method is employed to numerically solve the regularized long wave equations. Initially, the temporal derivative is discretized using the forward difference operator, leading to the development of a semi-discrete scheme in the time domain. Subsequently, a fully discrete formulation is obtained by discretizing the spatial variable using the Müntz spectral element method. The stability of the semi-discrete time scheme is analyzed and confirmed through Fourier analysis. Additionally, the convergence of the fully discrete scheme is rigorously established. Finally, by solving several numerical examples and comparing the results with outcomes from existing methods, the accuracy and efficiency of the Müntz spectral element method are effectively demonstrated.

1. Introduction

Modeling many problems in various disciplines of science and engineering leads to partial differential equations (PDEs). In most cases, these problems do not have exact solutions, or the exact solutions are complicated. The development of numerical methods to approximate the solutions to these problems has been very important and has always attracted the interest of mathematicians and engineers. In the following, we briefly introduce some numerical methods for solving PDEs.

Finite difference approximations for derivatives are among the simplest and oldest methods to solve differential equations. This method was already known by Euler in 1768 for one-dimensional problems, and was probably extended to two dimensions by Runge in 1908. The accuracy of the finite difference method (FDM) can be assessed by comparing the size of its truncation errors, which can be found by replacing the approximate solution with the exact solution [1].

The finite volume method (FVM) is a method for representing and evaluating partial differential equations in the form of algebraic equations [2]. In the finite volume method, volume integrals in a PDE that contain divergence terms are converted to surface integrals using the divergence theorem. These terms are then evaluated as fluxes at the surfaces of each finite volume. Because the flux entering a given volume is identical to that leaving the adjacent volume, these methods are conservative. Another advantage of the finite volume method is

its ease of formulation on unstructured meshes. This method is used in many computational fluid dynamics packages. The term “finite volume” refers to the small volume surrounding each node point on a mesh [3]. Finite volume methods can be compared and contrasted with finite difference methods, which approximate derivatives using nodal values, or finite element methods, which create local approximations of a solution using local data and construct a global approximation by stitching them together. In contrast, a finite volume method evaluates exact expressions for the average value of the solution over some volume, and uses this data to construct approximations of the solution within cells [4,5].

The spectral method is a conventional technique for solving PDEs, first introduced by Navier for elastic sheet problems in 1825. The main property of spectral methods is their high accuracy and rapid asymptotic rate of convergence, known as exponential convergence. Spectral methods can be divided into two general categories. First, collocation methods, which were initially introduced by Slater [6].

Second, non-interpolating methods include the Tau and Galerkin methods [7,8].

Another common method for solving PDEs is the finite element method (FEM). This method was first used to solve complex elasticity and structural analysis problems in civil and aeronautical engineering. The earliest works in this field were presented by Hrennikoff [9] and Courant [10] in the early 1940s.

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A rigorous mathematical foundation for FEM was provided in 1973 with the publication by Strang and Fix [11]. Since then, the method has been generalized for numerical modeling of physical systems in various engineering disciplines, including electromagnetism, heat transfer, and fluid dynamics [12,13]. The most important property of FEM is its flexibility in solving problems on unstructured domains.

In 1984, Patera introduced a new method, called the spectral element method (SEM), by combining the flexibility of low-order FEMs on unstructured domains with the high accuracy and exponential convergence of spectral methods [14]. Patera used Lagrange polynomials based on Gauss-Lobatto-Legendre points. Common polynomials used as basis functions in SEMs are Chebyshev and Legendre polynomials. The main advantage of using Legendre polynomials over Chebyshev polynomials is the diagonal mass matrix obtained in spectral element methods (LSEMs) [15]. The use of this method for solving various science and engineering problems has increased significantly in recent years.

Zhu et al. used the Chebyshev SEM coupled with the implicit Newmark time integration method to simulate acoustic fields [16]. Yibiao Li and Xin Kai Li developed a new Chebyshev SEM in their paper, where exact quadratures are used to overcome a shortcoming of the Gauss-Chebyshev quadrature in variational spectral element projections. They tested the method with the Stokes and the Cauchy-Riemann problems, showing that the numerical accuracy is much better than that of the Gauss-Lobatto-Legendre SEM [17]. Quaglino et al. used LSEMs for fast and accurate training of Neural Ordinary Differential Equations [18]. Lotfi and Alipanah solved the sine-Gordon equation in one dimension with LSEM [19]. In [20], the authors proposed a new numerical scheme based on LSEM for solving two- and three-dimensional time-fractional nonlinear damped Klein-Gordon equations. In [21], a reduced-order modal SEM was investigated to solve a two-dimensional viscoelastic equation. Dehghan et al. used LSEM to solve Sobolev equations that have applications in physics and mechanical engineering, and performed error analysis for the introduced method [22].

In this paper, we consider the regularized long-wave equation (RLWE), proposed by Benjamin et al. [23]. It is one of the well-known types of nonlinear evolution equations that play a fundamental role in mathematical physics and engineering fields such as chemical kinetics, solid-state physics, geochemistry, plasma physics, chemical physics, and fluid mechanics [24]. The RLWE, as a class of nonlinear evolution equations predicting a wide range of physical phenomena, was first introduced by Peregrine to model undular bore behavior [25], and is modeled as follows [26,27]:

$$\frac{\partial u}{\partial t} + (1 + \varepsilon u) \frac{\partial u}{\partial x} - \mu \frac{\partial}{\partial t} \left(\frac{\partial^2 u}{\partial x^2} \right) = 0, \quad 0 \leq t \leq T, \quad \alpha \leq x \leq \beta. \quad (1.1)$$

with the physical boundary conditions $u \rightarrow 0$ as $x \rightarrow \pm\infty$, where t is time, x is the spatial coordinate, $u(x, t)$ is the wave amplitude, and the positive parameters ε and μ are given. The subscripts t and x denote differentiation. Eq. (1.1) appears in modeling problems such as phonon packets in nonlinear crystals, magnetohydrodynamic and ion-acoustic waves in plasma, nonlinear transverse waves in shallow water, and others [28]. Although RLWE has analytical solutions under limited conditions, various numerical methods have been introduced to solve it.

Hassan et al. introduced the FDM for numerical solution of (1.1), using a combination of leapfrog for time dependence and pseudospectral (Fourier transform) treatment for space dependence [29]. The use of FDM is also suggested in [30–32]. Esen and Kutluay applied a lumped Galerkin method based on quadratic B-spline finite elements to find numerical solutions of the RLWE with various initial and boundary conditions [33]. In [34], the RLW and modified RLW equations were solved numerically using Bernstein polynomials in both space and time, based on the Kronecker product. The finite element method is another numerical technique for solving RLWE (1.1). In [35], a powerful numerical scheme was developed using the Petrov-Galerkin

method with cubic element shape functions and quadratic B-spline weight functions. Guo and Chen proposed an H^1 -Galerkin mixed FEM for the 1-D RLWE [36]. In [37], a new cubic B-spline least-squares finite-element method was presented. An adaptive moving mesh FEM for the numerical solution of RLWE was proposed by Lu et al. [38].

We consider the RLWE (1.1) with the boundary conditions

$$u(\alpha, t) = 0, \quad u(\beta, t) = 0, \quad t > 0, \quad (1.2)$$

and initial condition

$$u(x, 0) = f_1(x). \quad (1.3)$$

The framework of this paper is as follows: In Section 2, we introduce Müntz polynomials. In Section 3, the spectral element method (SEM) is defined based on Müntz polynomials. In Section 4, we apply the Müntz spectral element method (MSEM) to solve the regularized long-wave equation (RLWE). Section 5 includes the stability and convergence analysis of our method. Section 6 provides numerical examples using MSEM. Finally, the paper is concluded in Section 7.

2. Müntz polynomials

In 1885, Weierstrass introduced one of the most fundamental theorems in approximation theory:

Theorem 2.1 (Weierstrass Approximation Theorem). *Let f be a continuous real-valued function defined on the interval $[0, 1]$, and let $\varepsilon > 0$. Then there exists a polynomial p on $[0, 1]$ such that*

$$\|f - p\| < \varepsilon.$$

In other words, the set $\text{span}\{x^n : n \in \mathbb{Z}^+ \cup \{0\}\}$ is dense in $C([0, 1])$ with respect to the supremum norm. A natural question, raised by the great approximation theorist Bernstein, was how many exponents are needed for such density in $C([0, 1])$.

In 1912, Bernstein conjectured the following: given a sequence of distinct real numbers

$$\{0 = \lambda_0 < \lambda_1 < \lambda_2 < \dots\},$$

the set

$$\text{span}\{x^{\lambda_j} : j \in \mathbb{Z}^+ \cup \{0\}\}$$

is dense in $C([0, 1])$ if and only if

$$\sum_{j=1}^{\infty} \frac{1}{\lambda_j} = \infty.$$

Bernstein also proved necessary and sufficient conditions for density:

$$\sum_{j=1}^{\infty} \frac{1 + \log \lambda_j}{\lambda_j} = \infty,$$

and

$$\sum_{j=1}^{\infty} \frac{\lambda_j}{j \log \lambda_j} = 0.$$

Bernstein's conjecture was proved independently by Hermann Müntz in 1914 [39] and Otto Szász in 1916 [40].

Theorem 2.2 (Müntz-Szász). *Let $\{\lambda_j\}_{j=0}^{\infty}$ be a sequence of real numbers with*

$$0 = \lambda_0 < \lambda_1 < \lambda_2 < \dots \quad \text{and} \quad \lim_{j \rightarrow \infty} \lambda_j = \infty.$$

Then the set

$$\text{span}\{x^{\lambda_j} : j \in \mathbb{Z}^+ \cup \{0\}\}$$

is dense in $C([0, 1])$ if and only if

$$\sum_{j=1}^{\infty} \frac{1}{\lambda_j} = \infty.$$

Let $\{\lambda_j\}_{j=0}^\infty$ be as above. The classical Müntz–Szász theorem states that Müntz polynomials are linear combinations of the Müntz system

$$\{x^{\lambda_0}, x^{\lambda_1}, \dots, x^{\lambda_n}\},$$

which we denote by

$$M_n(\lambda) = \text{span}\{x^{\lambda_0}, x^{\lambda_1}, \dots, x^{\lambda_n}\}.$$

The Müntz orthogonal system was first introduced by Armenian mathematicians [41,42]. In this paper, we consider the Müntz–Legendre polynomials, which form an orthogonal system on the interval $[0, 1]$ with respect to the weight function $w(x) = 1$, and have been applied in numerical solutions of various problems [43–45].

2.1. Orthogonal Müntz–Legendre polynomials

Suppose in a sequence $\{\lambda_0, \lambda_1, \lambda_2, \dots\}$, we have $\lambda_k > 0$ for $k = 0, 1, 2, \dots$, and define $\Lambda_n = \{\lambda_0, \lambda_1, \dots, \lambda_n\}$.

If a simple contour Γ contains and surrounds all the zeros of the denominator in the following function

$$W_n(s) = \prod_{k=0}^{n-1} \frac{s + \bar{\lambda}_k + 1}{s - \lambda_k} \cdot \frac{1}{s - \lambda_n}, \quad n \in \mathbb{N} \cup \{0\},$$

where $\bar{\lambda}_k$ denotes the complex conjugate of λ_k (for complex λ_k) or simply λ_k itself (for real λ_k). We define the n th Müntz–Legendre polynomial as follows [41,42]:

$$P_n(x) = P_n(x; \Lambda_n) = \frac{1}{2\pi i} \oint_{\Gamma} W_n(s) x^s ds.$$

In the case $n = 0$, we consider $P_0(x) = 1$. If $\lambda_k \neq \lambda_l$ for $k \neq l$, these polynomials can be expressed as follows [46]:

$$P_n(x) = \sum_{k=0}^n C_{n,k} x^{\lambda_k}, \quad C_{n,k} = \begin{cases} 1 & \text{if } n = 0 \\ \frac{\prod_{l=0}^{n-1} (\lambda_k + \lambda_l + 1)}{\prod_{l=0, l \neq k}^n (\lambda_k - \lambda_l)} & \text{if } n \geq 1 \end{cases} \quad (2.1)$$

The Müntz–Legendre polynomials satisfy the relation [47]:

$$x P_n'(x) - x P_{n-1}'(x) = \lambda_n P_n(x) + (1 + \lambda_{n-1}) P_{n-1}(x). \quad (2.2)$$

They also satisfy the orthogonality condition [47]:

$$(P_n, P_m) = \int_0^1 P_n(x) \overline{P_m(x)} dx = \frac{\delta_{n,m}}{2\lambda_n + 1}. \quad (2.3)$$

The first four Müntz–Legendre polynomials for $\lambda_k = \frac{k}{5}$ are:

$$\begin{cases} P_0(x) = 1, \\ P_1(x) = 6x^{\frac{1}{5}} - 5, \\ P_2(x) = 28x^{\frac{2}{5}} - 42x^{\frac{1}{5}} + 15, \\ P_3(x) = 168x^{\frac{3}{5}} - 252x^{\frac{2}{5}} + 120x^{\frac{1}{5}} - 35. \end{cases}$$

Figs. 1 and 2 show the graphs of Müntz–Legendre polynomials P_n for different values of n and λ_k .

3. MSEM

To solve problems using the spectral element method (SEM), first, we divide the domain into N_e non-overlapping subdomains (elements) Ω_e , satisfying the following conditions [19]:

$$\bar{\Omega} = \bigcup_{e=1}^{N_e} \bar{\Omega}_e, \quad \Omega_e \cap \Omega_{e'} = \emptyset, \quad e \neq e'. \quad (3.1)$$

We define the approximation space U^h as

$$U^h = \{u \in U : u|_{\Omega_e} \in \mathbb{P}_N(\Omega_e)\}, \quad (3.2)$$

where $\mathbb{P}_N$ is the space of polynomials of degree less than or equal to N .

In spectral element methods, convergence speed can be improved by increasing the polynomial degree or the number of elements N_e . Basis

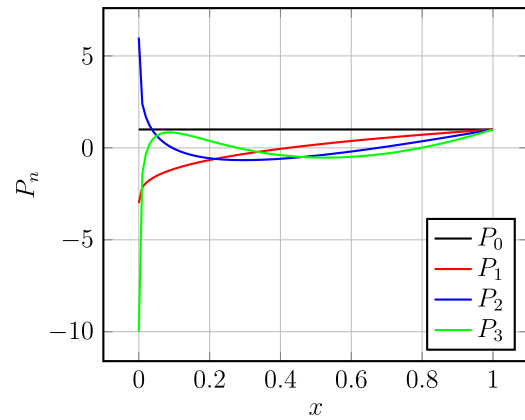


Fig. 1. Graph of the Müntz–Legendre polynomials with $n = 0, 1, 2, 3$ and $\lambda_k = \frac{k}{3}$.

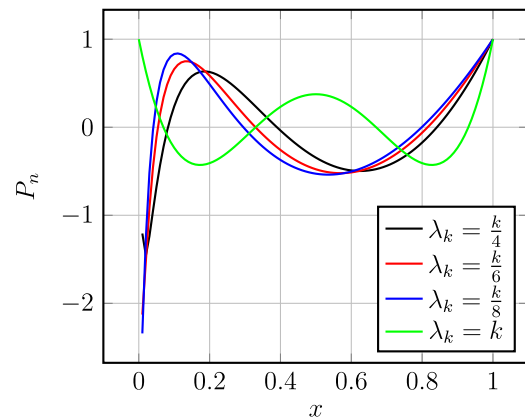


Fig. 2. Graph of the Müntz–Legendre polynomial P_4 for $\lambda_k = \frac{k}{4}, \frac{k}{6}, \frac{k}{8}, k$.

functions are typically chosen as high-degree Lagrangian interpolation polynomials defined at Gauss–Lobatto points on each element. Here, we introduce a new basis for the SEM using a special combination of Müntz polynomials.

3.1. MSEM

After dividing the domain into N_e non-overlapping elements, the approximate solution of order N , u^e , on each element is defined as follows:

$$u^e = \sum_{j=0}^{N-2} u_j^e \varphi_j, \quad 1 \leq e \leq N_e, \quad (3.3)$$

where the basis functions φ_j are defined as

$$\varphi_j = Q_j - Q_{j+2}, \quad j = 0, 1, \dots, N-2. \quad (3.4)$$

Here, Q_j is a combination of Müntz–Legendre polynomials (2.1) with $\lambda_k = k$, $k = 0, \dots, j$ mapped to the interval $[-1, 1]$, defined as follows:

$$Q_j(x) = P_j\left(\frac{x+1}{2}\right) = \sum_{k=0}^j C_{j,k} \left(\frac{x+1}{2}\right)^k, \quad C_{j,k} = \frac{(-1)^{j+k}}{k!(j-k)!} \prod_{l=0}^{j-1} (k+l+1). \quad (3.5)$$

Fig. 3 shows the graph of φ_j for $j = 0, 1, 2, 3, 4$.

Let t_e and t_{e-1} be the endpoints of the e th element. For the transformation between the interval $[-1, 1]$ and the e th element, we use the

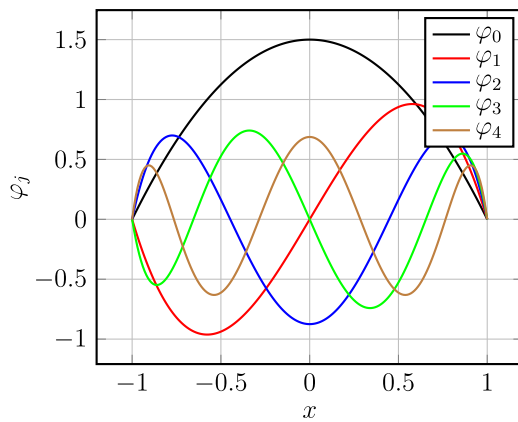


Fig. 3. Graph of the basis functions φ_j for $j = 0, 1, 2, 3, 4$.

following mapping functions:

$$\begin{cases} t(x) = \frac{(t_e - t_{e-1})}{2}x + \frac{t_e + t_{e-1}}{2}, & -1 \leq x \leq 1, \\ x(t) = \frac{2t - (t_e + t_{e-1})}{t_e - t_{e-1}}, & t_{e-1} \leq t \leq t_e. \end{cases}$$

The mass matrix

If $h_e = t_e - t_{e-1}$, then the elemental mass matrix M_{ij}^e on Ω_e is constructed as follows for $i, j = 0, 1, \dots, N-2$:

$$M_{ij}^e = \int_{t_{e-1}}^{t_e} \varphi_i(t) \varphi_j(t) dt = \frac{h_e}{2} \int_{-1}^1 \varphi_i(x) \varphi_j(x) dx.$$

The mass matrix is symmetric and matrix with three non-zero diagonals with entries

$$M_{ij}^e = \begin{cases} h_e \left(\frac{1}{2j+1} + \frac{1}{2j+5} \right), & i = j, \\ -h_e \frac{1}{2j+5}, & i = j+2, \\ -h_e \frac{1}{2j+1}, & i = j-2, \\ 0, & \text{otherwise.} \end{cases} \quad (3.6)$$

Remark 3.1. Unlike classical spectral methods, our mass matrix has non-zero entries on the main diagonal and the ± 2 diagonals due to the unique recurrence relations of Müntz-Legendre polynomials. The explicit mathematical justification:

$$M_{ij}^e = 0 \quad \text{for } |i-j| \neq 0, 2 \quad (3.7)$$

shows the exact non-zero pattern.

The stiffness matrix

Let $h_e = t_e - t_{e-1}$, then the stiffness matrix on each element is calculated as follows (for $i, j = 0, 1, \dots, N-2$):

$$S_{ij}^e = \int_{t_{e-1}}^{t_e} \frac{d\varphi_i(t)}{dt} \frac{d\varphi_j(t)}{dt} dt = \frac{2}{h_e} \int_{-1}^1 \frac{d\varphi_i(x)}{dx} \frac{d\varphi_j(x)}{dx} dx,$$

which is a diagonal matrix with entries

$$S_{ij}^e = \begin{cases} h_e^{-1}(8i+12), & i = j, \\ 0, & i \neq j. \end{cases} \quad (3.8)$$

The matrix with three non-zero diagonals form of the mass matrix $\mathbf{M}_e$ and the diagonal form of the stiffness matrix $\mathbf{S}_e$ directly result from the orthogonality properties of Müntz-Legendre polynomials and their recurrence relations. Specifically [48]:

For the basis functions $Q_j(x)$, the inner product $\langle Q_i, Q_j \rangle$ is nonzero only when $|i-j| \leq 1$, which yields the matrix with three non-zero diagonals structure of $\mathbf{M}_e$.

Table 1

Comparison of the condition number of mass and stiffness matrices in the two methods (MSEM and LSEM with $N = 5$)

Method	Mass	Stiffness
MSEM	2.52	1.66
LSEM	8.3229	9.64×10^{16}

The stiffness matrix $\mathbf{S}_e$ becomes diagonal due to the weighted orthogonality relation between the derivatives $Q'_j(x)$.

These properties have been numerically verified in MATLAB for various values of n .

Table 1 shows the condition number of mass and stiffness matrices in two methods, Legendre SEM (LSEM) and MSEM, for $N = 5$. According to the data in Table 1, the condition number of the mass matrix in the two methods is almost the same, but the condition number of the stiffness matrix of LSEM is much larger than the corresponding matrix in MSEM.

4. MSEM for RLWE

4.1. Time discrete scheme

To obtain a semi-discrete scheme in time for Eq. (1.1), first divide the interval $[0, T]$ into M subintervals $[t_m, t_{m+1}]$, where $t_{m+1} = t_m + \tau$, $t_0 = 0$, and $m = 0, 1, \dots, M-1$. Using the forward difference operator for the first time derivative and central difference operator for u^n , and locally freezing the nonlinear term $u^n u_x^n = \varpi u_{xx}^n$ [28,33,49], we have

$$\frac{u^{n+1} - u^n}{\tau} + \frac{u_x^n + u_x^{n+1}}{2} + \varepsilon \varpi \frac{u_x^n + u_x^{n+1}}{2} - \mu \frac{u_{xx}^{n+1} - u_{xx}^n}{\tau} = 0.$$

Multiplying the above equation by τ , we obtain

$$u^{n+1} + \tau \left(\frac{1 + \varepsilon \varpi}{2} \right) u_x^{n+1} - \mu u_{xx}^{n+1} = u^n - \tau \left(\frac{1 + \varepsilon \varpi}{2} \right) u_x^n - \mu u_{xx}^n. \quad (4.1)$$

4.2. Space discrete scheme

The weak form of Eq. (4.1) is as follows: for each element Ω_e , find $u^e \in U^h$ such that

$$\begin{aligned} & \int_{\Omega_e} \left(u^{n+1} + \tau \frac{1 + \varepsilon \varpi}{2} u_x^{n+1} - \mu u_{xx}^{n+1} \right) v \, dx \\ &= \int_{\Omega_e} \left(u^n - \tau \frac{1 + \varepsilon \varpi}{2} u_x^n - \mu u_{xx}^n \right) v \, dx, \quad \forall v \in U^h, \quad 1 \leq e \leq N_e. \end{aligned}$$

By integration by parts for terms involving the second spatial derivative, we have

$$\begin{aligned} & \int_{\Omega_e} u^{n+1} v \, dx + \tau \frac{1 + \varepsilon \varpi}{2} \int_{\Omega_e} u_x^{n+1} v \, dx + \mu \left(u_x^{n+1} v \Big|_{\partial \Omega_e} - \int_{\Omega_e} u_x^{n+1} v_x \, dx \right) \\ &= \int_{\Omega_e} u^n v \, dx - \tau \frac{1 + \varepsilon \varpi}{2} \int_{\Omega_e} u_x^n v \, dx + \mu \left(u_x^n v \Big|_{\partial \Omega_e} - \int_{\Omega_e} u_x^n v_x \, dx \right). \end{aligned}$$

To apply the method and implement it computationally, the solution space is restricted to a finite subspace $V^h \subset H_0^1$ with basis functions $\{\varphi_i\}_{i=0}^{N-2}$ defined in (3.4), and we look for an approximate solution in this subspace as

$$w^n(x) = \sum_{i=0}^{N-2} u_i^n \varphi_i(x). \quad (4.2)$$

The solution coefficients u_i^n can be calculated by solving the following equation:

$$\begin{aligned} & (w^{n+1}, v) + \tau \frac{1 + \varepsilon \varpi}{2} (w_x^{n+1}, v) + \mu \left(w_x^{n+1} v \Big|_{\partial \Omega_e} - (w_x^{n+1}, v_x) \right) \\ &= (w^n, v) - \tau \frac{1 + \varepsilon \varpi}{2} (w_x^n, v) + \mu \left(w_x^n v \Big|_{\partial \Omega_e} - (w_x^n, v_x) \right), \quad \forall v \in V^h. \end{aligned} \quad (4.3)$$

By substituting (4.2) into (4.3) for w^k , $k = n, n+1$, and using the basis functions defined in (3.4) and (3.5), noting that $w_x^n v|_{\partial\Omega_e} = w_x^{n+1} v|_{\partial\Omega_e} = 0$, we obtain

$$\begin{aligned} & \sum_{i=0}^{N-2} u_i^{n+1} \left(\int_{\Omega_e} \varphi_i \varphi_j dx + \tau \frac{1+\varepsilon\varpi}{2} \int_{\Omega_e} \varphi'_i \varphi_j dx - \mu \int_{\Omega_e} \varphi'_i \varphi'_j dx \right) \\ &= \sum_{i=0}^{N-2} u_i^n \left(\int_{\Omega_e} \varphi_i \varphi_j dx + \tau \frac{1+\varepsilon\varpi}{2} \int_{\Omega_e} \varphi'_i \varphi_j dx - \mu \int_{\Omega_e} \varphi'_i \varphi'_j dx \right). \end{aligned}$$

The matrix form of the above equation is

$$\left(M^e + \tau \frac{1+\varepsilon\varpi}{2} A^e - \mu S^e \right) U^{n+1} = \left(M^e - \tau \frac{1+\varepsilon\varpi}{2} A^e - \mu S^e \right) U^n, \quad (4.4)$$

where

$$M_{ij}^e = (\varphi_i, \varphi_j), \quad S_{ij}^e = (\varphi'_i, \varphi'_j), \quad A_{ij}^e = (\varphi'_i, \varphi_j). \quad (4.5)$$

Remark 4.1. The values of the matrix A^e in (4.5) are calculated as

$$A_{ij}^e = \begin{cases} 2, & i = j + 1, \\ -2, & i = j - 1, \\ 0, & \text{otherwise.} \end{cases} \quad (4.6)$$

In other words, $A = \text{trid}(2, 0, -2)$.

The matrix A_e exhibits an extended matrix with three non-zero diagonals structure (with non-zero entries along the ± 2 diagonals) due to the second-order differential operator and the product properties of the basis functions $Q_j(x)$. Specifically:

This operator couples basis functions separated by two degrees, resulting in the aforementioned sparsity pattern.

The validity of this structure has been confirmed both analytically and numerically for various values of n .

5. Stability and convergence analysis

In this section, we analyze the stability of (4.1) and the convergence of the MSEM presented in the previous sections.

5.1. Stability of time discretization

To investigate the stability of the scheme (4.1), we use the Fourier method based on the Von Neumann theory [50]. To this end, we replace the Fourier mode $u^n = \xi^n e^{i\beta x}$ into (4.1). Then, we have

$$\begin{aligned} & \xi^{n+1} e^{i\beta x} + \tau \left(\frac{1+\varepsilon\varpi}{2} \right) (i\beta) \xi^{n+1} e^{i\beta x} - \mu (i\beta)^2 \xi^{n+1} e^{i\beta x} \\ &= \xi^n e^{i\beta x} - \tau \left(\frac{1+\varepsilon\varpi}{2} \right) (i\beta) \xi^n e^{i\beta x} - \mu (i\beta)^2 \xi^n e^{i\beta x}. \end{aligned}$$

By dividing the above equation by $\xi^n e^{i\beta x}$, we obtain

$$\xi + \tau \left(\frac{1+\varepsilon\varpi}{2} \right) (i\beta) \xi - \mu (i\beta)^2 \xi = 1 - \tau \left(\frac{1+\varepsilon\varpi}{2} \right) (i\beta) - \mu (i\beta)^2.$$

Rearranging gives

$$\xi = \frac{1 - \tau \left(\frac{1+\varepsilon\varpi}{2} \right) (i\beta) - \mu (i\beta)^2}{1 + \tau \left(\frac{1+\varepsilon\varpi}{2} \right) (i\beta) - \mu (i\beta)^2}.$$

Since $i^2 = -1$, we simplify to

$$\xi = \frac{1 - \tau \left(\frac{1+\varepsilon\varpi}{2} \right) (i\beta) + \mu \beta^2}{1 + \tau \left(\frac{1+\varepsilon\varpi}{2} \right) (i\beta) + \mu \beta^2},$$

which can be rewritten as

$$\xi = \frac{a - bi}{a + bi}, \quad a = 1 + \mu \beta^2, \quad b = \tau \left(\frac{1+\varepsilon\varpi}{2} \right) \beta. \quad (5.1)$$

Taking the modulus of the growth factor gives $|\xi| = 1$, and so the scheme (4.1) is unconditionally stable [35].

5.2. Convergence of MSEM

Here, we obtain an error bound for the proposed numerical scheme. So, if we define the spatial projection operator

$$\mathcal{L}_N^1 : H_0^1(\Omega) \rightarrow \mathfrak{N}_N^0,$$

where $H_0^1(\Omega) = \{u \in H^1(\Omega) : \mathfrak{T}u = 0 \text{ on } \partial\Omega\}$, then it is clear that [8]

$$(\nabla(u - \mathcal{L}_N^1 u), \nabla v) = 0, \quad u \in H_0^1(\Omega), \quad \forall v \in \mathfrak{N}_N^0. \quad (5.2)$$

$\mathfrak{N}_N^0$ is the SEM approximation space

$$\mathfrak{N}_N^0 = \{w \in H_0^1(\Omega) : w|_{\Omega_e} \in \mathbb{P}_N(\Omega), \quad e = 1, \dots, N_e\}.$$

Lemma 5.1. If $u \in H^\kappa$, with $\kappa \geq 1$, then [8]

$$\|u - \mathcal{L}_N^1 u\| \leq C \left[\sum_{j=1}^{N_e} h_j^{2(\min(N_j+1, \kappa)-1)} N_j^{2(1-\kappa)} \|u\|_\kappa^2 \right]^{\frac{1}{2}}. \quad (5.3)$$

In the special case where $N_j = N$ and $h \leq h_j \leq c'h$, the following inequality is obtained

$$\|u - \mathcal{L}_N^1 u\| \leq C h_j^{\min(N+1, \kappa)-1} N^{1-\kappa} \|u\|_\kappa. \quad (5.4)$$

Theorem 5.2. For non-negative sequences a_n , b_n , and c_n , if

$$c_0 \leq \chi_0, \quad c_n \leq \chi_0 + \sum_{i=0}^n b_i + \sum_{i=0}^n a_i c_i,$$

then [51]

$$c_n \leq \left(\chi_0 + \sum_{i=0}^n b_i \right) e^{\sum_{i=0}^n a_i},$$

where $\chi_0 \geq 0$.

Definition 5.3. Let $u^n = u(x, t_n)$ be the exact solution of Eqs. (1.1)–(1.3) and w^n be the solution of the full discrete scheme (4.3). We define the following notations

$$Y_N^n = u^n - \mathcal{L}_N^1 u^n, \quad \mathfrak{Y}_N^n = \mathcal{L}_N^1 u^n - w^n. \quad (5.5)$$

Lemma 5.4. Let $\mathcal{Y}_N^{n+1} = Y_N^{n+1} - Y_N^n$, then

$$\|\mathcal{Y}_N^{n+1}\| \leq C \tau h_j^{\min(N+1, \kappa)-1} N^{1-\kappa} \|u\|_\kappa. \quad (5.6)$$

Proof.

$$\begin{aligned} \|\mathcal{Y}_N^{n+1}\| &= \|Y_N^{n+1} - Y_N^n\| = \|u^{n+1} - u^n - (\mathcal{L}_N^1 u^{n+1} - \mathcal{L}_N^1 u^n)\| \\ &= \left\| \int_{t_n}^{t_{n+1}} \frac{\partial u}{\partial t} dt - \int_{t_n}^{t_{n+1}} \frac{\partial \mathcal{L}_N^1 u}{\partial t} dt \right\| = \left\| \int_{t_n}^{t_{n+1}} \left(\frac{\partial u}{\partial t} - \frac{\partial \mathcal{L}_N^1 u}{\partial t} \right) dt \right\| \\ &\leq \int_{t_n}^{t_{n+1}} \left\| \frac{\partial u}{\partial t} - \frac{\partial \mathcal{L}_N^1 u}{\partial t} \right\| dt. \end{aligned} \quad (5.7)$$

Now, using (5.4) we have

$$\begin{aligned} \int_{t_n}^{t_{n+1}} \left\| \frac{\partial u}{\partial t} - \frac{\partial \mathcal{L}_N^1 u}{\partial t} \right\| dt &= \int_{t_n}^{t_{n+1}} \left\| \frac{\partial}{\partial t} (u - \mathcal{L}_N^1 u) \right\| dt \\ &\leq \int_{t_n}^{t_{n+1}} C h_j^{\min(N+1, \kappa)-1} N^{1-\kappa} dt \\ &= C h_j^{\min(N+1, \kappa)-1} \tau N^{1-\kappa} \|u\|_\kappa. \end{aligned} \quad (5.8)$$

From (5.7) and (5.8), we have

$$\|\mathcal{Y}_N^{n+1}\| \leq C \tau h_j^{\min(N+1, \kappa)-1} N^{1-\kappa} \|u\|_\kappa. \quad \square$$

Lemma 5.5. From the notations (5.5), we have

$$\|\mathfrak{I}_N^{n+1}\|_{L^2(\Omega)}^2 \leq \|\mathfrak{I}_N^0\|_{L^2(\Omega)}^2 + (n+1)\vartheta C\tau N^{2-2\kappa} e^{(n+1)\vartheta}.$$

Proof. From Eq. (4.3), we have

$$\begin{aligned} (u^{n+1}, v) + \tau \left(\frac{1+\varepsilon\varpi}{2} \right) (u_x^{n+1}, v) - \mu(u_x^{n+1}, v_x) \\ = (u^n, v) - \tau \left(\frac{1+\varepsilon\varpi}{2} \right) (u_x^n, v) - \mu(u_x^n, v_x). \end{aligned} \quad (5.9)$$

Subtracting Eq. (5.9) from Eq. (4.3) yields

$$\begin{aligned} (u^{n+1} - w^{n+1}, v) + c_1(u_x^{n+1} - w_x^{n+1}, v) - c_2(u_x^{n+1} - w_x^{n+1}, v_x) \\ = (u^n - w^n, v) - c_1(u_x^n - w_x^n, v) - c_2(u_x^n - w_x^n, v_x), \end{aligned} \quad (5.10)$$

where $c_1 = \tau \frac{1+\varepsilon\varpi}{2}$ and $c_2 = \mu$. Now, using the projection operator $\mathfrak{L}_N^1$ we have

$$\begin{aligned} (u^{n+1} - \mathfrak{L}_N^1 u^{n+1} + \mathfrak{L}_N^1 u^{n+1} - w^{n+1}, v) \\ + c_1 \left(\frac{\partial}{\partial x} (u^{n+1} - \mathfrak{L}_N^1 u^{n+1} + \mathfrak{L}_N^1 u^{n+1} - w^{n+1}), v \right) \\ - c_2 \left(\frac{\partial}{\partial x} (u^{n+1} - \mathfrak{L}_N^1 u^{n+1} + \mathfrak{L}_N^1 u^{n+1} - w^{n+1}), v_x \right) \\ = (u^n - \mathfrak{L}_N^1 u^n + \mathfrak{L}_N^1 u^n - w^n, v) - c_1 \left(\frac{\partial}{\partial x} (u^n - \mathfrak{L}_N^1 u^n + \mathfrak{L}_N^1 u^n - w^n), v \right) \\ - c_2 \left(\frac{\partial}{\partial x} (u^n - \mathfrak{L}_N^1 u^n + \mathfrak{L}_N^1 u^n - w^n), v_x \right), \end{aligned} \quad (5.11)$$

So we have

$$\begin{aligned} (\mathfrak{I}_N^{n+1}, v) + c_1 \left(\frac{\partial}{\partial x} (\mathfrak{I}_N^{n+1}), v \right) - c_2 \left(\frac{\partial}{\partial x} (\mathfrak{I}_N^{n+1}), v_x \right) \\ = (\mathcal{Y}_N^n + \mathfrak{I}_N^n, v) - c_1 \left(\frac{\partial}{\partial x} (\mathcal{Y}_N^n + \mathfrak{I}_N^n), v \right) \\ - c_2 \left(\frac{\partial}{\partial x} (\mathcal{Y}_N^n + \mathfrak{I}_N^n), v_x \right) - (\mathcal{Y}_N^{n+1}, v) \\ - c_1 \left(\frac{\partial}{\partial x} (\mathcal{Y}_N^{n+1}), v \right) + c_2 \left(\frac{\partial}{\partial x} (\mathcal{Y}_N^{n+1}), v_x \right), \end{aligned} \quad (5.12)$$

We set $v = \mathfrak{I}_N^{n+1}$, which yields

$$\begin{aligned} (\mathfrak{I}_N^{n+1}, \mathfrak{I}_N^{n+1}) - c_2 \left(\frac{\partial}{\partial x} (\mathfrak{I}_N^{n+1}), \frac{\partial}{\partial x} (\mathfrak{I}_N^{n+1}) \right) \\ = (\mathfrak{I}_N^n, \mathfrak{I}_N^{n+1}) - c_1 \left(\frac{\partial}{\partial x} (\mathfrak{I}_N^n), \mathfrak{I}_N^{n+1} \right) \\ - c_1 \left(\frac{\partial}{\partial x} (\mathfrak{I}_N^{n+1}), \mathfrak{I}_N^{n+1} \right) - c_2 \left(\frac{\partial}{\partial x} (\mathfrak{I}_N^n), \frac{\partial}{\partial x} (\mathfrak{I}_N^{n+1}) \right) \\ + (\mathcal{I}_N^{n+1}, \mathfrak{I}_N^{n+1}) + c_1 \left(\frac{\partial}{\partial x} (\mathcal{I}_N^{n+1}), \mathfrak{I}_N^{n+1} \right) + c_2 \left(\frac{\partial}{\partial x} (\mathcal{I}_N^{n+1}), \frac{\partial}{\partial x} (\mathfrak{I}_N^{n+1}) \right). \end{aligned} \quad (5.13)$$

Applying Schwartz and Young's inequalities lead to

$$\begin{aligned} \|\mathfrak{I}_N^{n+1}\|_{L^2\Omega}^2 + c_2 \left\| \frac{\partial}{\partial x} (\mathfrak{I}_N^{n+1}) \right\|_{L^2\Omega}^2 \\ \leq \frac{1}{2} \|\mathfrak{I}_N^n\|_{L^2\Omega}^2 + \frac{1}{2} \|\mathfrak{I}_N^{n+1}\|_{L^2\Omega}^2 + \frac{c_1}{2} \left\| \frac{\partial}{\partial x} (\mathfrak{I}_N^n) \right\|_{L^2\Omega}^2 + \frac{c_1}{2} \|\mathfrak{I}_N^{n+1}\|_{L^2\Omega}^2 \\ + \frac{c_1}{2} \left\| \frac{\partial}{\partial x} (\mathfrak{I}_N^n) \right\|_{L^2\Omega}^2 + \frac{c_1}{2} \|\mathfrak{I}_N^{n+1}\|_{L^2\Omega}^2 + \frac{c_2}{2} \left\| \frac{\partial}{\partial x} (\mathfrak{I}_N^n) \right\|_{L^2\Omega}^2 \\ + \frac{c_2}{2} \left\| \frac{\partial}{\partial x} (\mathfrak{I}_N^{n+1}) \right\|_{L^2\Omega}^2 + \frac{1}{2} \|\mathcal{I}_N^{n+1}\|_{L^2\Omega}^2 + \frac{1}{2} \|\mathfrak{I}_N^{n+1}\|_{L^2\Omega}^2 + \frac{c_1}{2} \left\| \frac{\partial}{\partial x} (\mathcal{I}_N^{n+1}) \right\|_{L^2\Omega}^2 \\ + \frac{c_1}{2} \|\mathfrak{I}_N^{n+1}\|_{L^2\Omega}^2 + \frac{c_2}{2} \left\| \frac{\partial}{\partial x} (\mathcal{I}_N^{n+1}) \right\|_{L^2\Omega}^2 + \frac{c_2}{2} \left\| \frac{\partial}{\partial x} (\mathfrak{I}_N^{n+1}) \right\|_{L^2\Omega}^2. \end{aligned} \quad (5.14)$$

The above relation can be written to the following form

$$\begin{aligned} \left(\frac{1-3c_1}{2} \right) \|\mathfrak{I}_N^{n+1}\|_{L^2\Omega}^2 \leq \frac{1}{2} \|\mathfrak{I}_N^n\|_{L^2\Omega}^2 + \frac{2c_1+c_2}{2} \left\| \frac{\partial}{\partial x} (\mathfrak{I}_N^n) \right\|_{L^2\Omega}^2 \\ + \frac{1}{2} \|\mathcal{I}_N^{n+1}\|_{L^2\Omega}^2 + \frac{c_1+c_2}{2} \left\| \frac{\partial}{\partial x} (\mathcal{I}_N^{n+1}) \right\|_{L^2\Omega}^2. \end{aligned} \quad (5.15)$$

By taking the summation from (5.15), we get

$$\begin{aligned} \sum_{i=0}^n \left(\|\mathfrak{I}_N^{i+1}\|_{L^2\Omega}^2 - \|\mathfrak{I}_N^i\|_{L^2\Omega}^2 \right) \leq 3c_1 \sum_{i=0}^n \|\mathfrak{I}_N^{i+1}\|_{L^2\Omega}^2 \\ + (2c_1 + c_2) \sum_{i=0}^n \left\| \frac{\partial}{\partial x} (\mathfrak{I}_N^i) \right\|_{L^2\Omega}^2 \\ + \sum_{i=0}^n \|\mathcal{I}_N^{i+1}\|_{L^2\Omega}^2 + (c_1 + c_2) \sum_{i=0}^n \left\| \frac{\partial}{\partial x} (\mathcal{I}_N^{i+1}) \right\|_{L^2\Omega}^2. \end{aligned} \quad (5.16)$$

By using the telescoping series we have

$$\begin{aligned} \|\mathfrak{I}_N^{n+1}\|_{L^2\Omega}^2 \leq \|\mathfrak{I}_N^0\|_{L^2\Omega}^2 + 3c_1 \sum_{i=0}^n \|\mathfrak{I}_N^{i+1}\|_{L^2\Omega}^2 \\ + (2c_1 + c_2) \sum_{i=0}^n \left\| \frac{\partial}{\partial x} (\mathfrak{I}_N^i) \right\|_{L^2\Omega}^2 \\ + \sum_{i=0}^n \|\mathcal{I}_N^{i+1}\|_{L^2\Omega}^2 + (c_1 + c_2) \sum_{i=0}^n \left\| \frac{\partial}{\partial x} (\mathcal{I}_N^{i+1}) \right\|_{L^2\Omega}^2. \end{aligned} \quad (5.17)$$

According to Theorem 5.2, the following relations are achieved

$$\begin{aligned} \|\mathfrak{I}_N^{n+1}\|_{L^2\Omega}^2 \leq \|\mathfrak{I}_N^0\|_{L^2\Omega}^2 \\ + \vartheta \left(\sum_{i=0}^n \left(\|\mathcal{I}_N^{i+1}\|_{L^2\Omega}^2 + \left\| \frac{\partial}{\partial x} (\mathcal{I}_N^{i+1}) \right\|_{L^2\Omega}^2 \right) + \left(\|\mathfrak{I}_N^{n+1}\|_{L^2\Omega}^2 + \left\| \frac{\partial}{\partial x} (\mathfrak{I}_N^n) \right\|_{L^2\Omega}^2 \right) \right) \\ = \|\mathfrak{I}_N^0\|_{L^2\Omega}^2 + \vartheta \left(\sum_{i=0}^n \left(\|\mathcal{I}_N^{i+1}\|_{L^2\Omega}^2 + \left\| \frac{\partial}{\partial x} (\mathcal{I}_N^{i+1}) \right\|_{L^2\Omega}^2 \right) \right) e^{(n+1)\vartheta}, \end{aligned} \quad (5.18)$$

where $\vartheta = \max \{3c_1, 2c_1 + c_2, c_1 + c_2\}$.

Using Lemma 5.4, we can get

$$\begin{aligned} \|\mathfrak{I}_N^{n+1}\|_{L^2\Omega}^2 \leq \|\mathfrak{I}_N^0\|_{L^2\Omega}^2 + \vartheta \left(\sum_{i=0}^n (C\tau N^{2-2\kappa}) \right) e^{(n+1)\vartheta} \\ = \|\mathfrak{I}_N^0\|_{L^2\Omega}^2 + (n+1)\vartheta C\tau N^{2-2\kappa} e^{(n+1)\vartheta}. \end{aligned} \quad \square \quad (5.19)$$

Theorem 5.6. If u^n be the exact solution of Eqs. (1.1)–(1.3) and w^n be the solution of the full discrete scheme (4.3). Then, the following error estimation for the fully discrete scheme (4.3), will be obtained

$$\|u^n - w^n\| \leq \|\mathfrak{I}_N^0\| + C\tau N^{1-\kappa} \left(h_j^{(\min(N+1, \kappa)-1)} \|u\|_\kappa + (n+1)\vartheta e^{(n+1)\vartheta} \right).$$

Proof. By considering Definition 5.3 and Lemma 5.5, we get

$$\begin{aligned} \|u^n - w^n\| = \|\mathcal{Y}_N^{n+1} - \mathfrak{I}_N^{n+1}\| \leq \|\mathcal{Y}_N^{n+1}\| + \|\mathfrak{I}_N^{n+1}\| \\ \leq \|\mathfrak{I}_N^0\| + C\tau N^{1-\kappa} \left(h_j^{(\min(N+1, \kappa)-1)} \|u\|_\kappa + (n+1)\vartheta e^{(n+1)\vartheta} \right) \quad \square \end{aligned} \quad (5.20)$$

6. Illustrative examples

In this section we consider several numerical examples to show the accuracy and efficiency of the MSEM. We test the accuracy and stability of the MSEM described in this paper by performing the mentioned scheme for different values of N_e , N , and τ .

The RLWE (1.1) satisfies only three conservation laws given as [52]

$$\begin{aligned} I_1 &= \int_{-\infty}^{+\infty} u dx, \\ I_2 &= \int_{-\infty}^{+\infty} (u^2 + \mu(u_x)^2) dx, \\ I_3 &= \int_{-\infty}^{+\infty} (u^3 + 3u^2) dx. \end{aligned} \quad (6.1)$$

Table 2

Error norms for (MSSW) with different value of N and N_e , ($c = 0.1, 0.3, \tau = 0.1$ and $t = 20$).

N	N_e	$c = 0.1$		$c = 0.3$	
		L_2	L_∞	L_2	L_∞
3	10	1.3506×10^{-3}	6.2047×10^{-4}	3.0245×10^{-3}	2.1786×10^{-4}
4	9	7.2534×10^{-4}	1.8753×10^{-4}	9.2506×10^{-4}	1.0502×10^{-4}
5	8	2.0982×10^{-4}	7.1348×10^{-5}	3.0105×10^{-4}	8.8147×10^{-5}
6	7	8.9058×10^{-5}	4.1270×10^{-5}	9.6324×10^{-5}	6.2785×10^{-5}
7	6	7.5469×10^{-5}	3.9575×10^{-5}	7.1576×10^{-5}	4.9706×10^{-5}
8	5	9.4854×10^{-5}	3.8003×10^{-5}	8.1419×10^{-5}	6.4218×10^{-5}

Table 3

Comparison of error norms and invariants between the MSEM for (MSSW) with $N = 8$ and $N_e = 10$, and some methods in [29,37,55–58] for $c = 0.1, t = 20$ and $\tau = 0.1$.

	L_2	L_∞	I_1	I_2	I_3
Analytical			3.9799497	0.8104625	2.5790074
MSEM	6.8147×10^{-6}	2.1270×10^{-6}	3.9788622	0.8098939	2.5797930
[37]	–	1.645×10^{-4}	3.9820790	0.8112067	2.5814635
[55]	3.02×10^{-4}	1.15×10^{-4}	3.97988	0.81046	2.57900
[29]	2.3083×10^{-3}	6.656×10^{-4}	3.976063	0.809699	2.576489
[56]	–	7.344×10^{-5}	3.9798879	0.8104622	2.5790063
[57]	–	2.8834×10^{-5}	3.979962	0.810462	2.579007
[58]	–	9.5×10^{-6}	3.97988	0.81046	2.57900

We will use the L_2 and L_∞ error norms defined by

$$L_2 = \|u - w\|_2 = \left(h \sum_{i=0}^n (u^i - w^i)^2 \right)^{\frac{1}{2}},$$

$$L_\infty = \|u - w\|_\infty = \max_{0 \leq i \leq N} |u^i - w^i|.$$

6.1. The motion of a single solitary wave (MSSW)

In this case, we consider Eq. (1.1) with the following boundary and initial conditions

$$u(\alpha, t) = 0, u(\beta, t) = 0,$$

$$u(x, 0) = 3c \operatorname{sech}^2(p(x - x_0)),$$

where $p = \frac{1}{2} \left(\frac{\epsilon c}{\mu(1+\epsilon c)} \right)$. We use parameters $\epsilon = 1, \mu = 1$ and $x_0 = 0$. The exact solution is [53]

$$u(x, t) = 3c \operatorname{sech}^2(p(x - x_0 - (1 + \epsilon c)t)).$$

The analytical values of the invariants can be calculated as follows [54]

$$I_1 = \frac{6c}{p},$$

$$I_2 = \frac{12c^2}{p} + \frac{48pc^2\mu}{5},$$

$$I_3 = \frac{36c^2}{p} + \frac{144c^3}{5p}.$$

We numerically solve using the MSEM on the time interval $0 \leq t \leq 20$ and the space interval $-60 \leq x \leq 40$. Table 2 shows the comparison of error norms for different value of N and N_e , with $c = 0.1, 0.3, \tau = 0.1$ and $t = 20$. Table 3 shows the comparison of MSEM with $N = 8$ and $N_e = 10$ ($c = 0.1, \tau = 0.1$ and $t = 20$), already existing in the literatures (B-spline least-squares finite-element method [37], Haar Wavelet Method [55], fourier leapfrog method [29], Galerkin method [56], Operator splitting method [57] and a strang splitting approach combined with chebyshev wavelets [58]). It is clearly seen from this table that the MSEM gives satisfactory results for error norms when compared with the other ones. Also, invariants are in good agreement with the analytical results and results from the other methods. In Figs. 4 and 5, the propagation behaviors and absolute error are displayed in $0 \leq t \leq 20$ and $-60 \leq x \leq 40$, respectively. We also plotted the

Table 4

Comparison of invariants between the MSEM for wave undulation with $N = 8$ and $N_e = 10$, and some methods in [33,37,55,59] for $d = 2, t = 250$ and $\tau = 0.1$.

Method	$d = 2$		
	I_1	I_2	I_3
MSEM	30.61475	3.15058	9.71270
[37]	30.86531	3.18296	9.87011
[55]	30.475001	3.100580	9.609708
[59]	30.48700	3.10148	9.61203
[33]	30.4869971	3.10123	9.61118

Table 5

Comparison of invariants between the MSEM for wave undulation with $N = 8$ and $N_e = 10$, and some methods in [33,37,55,59] for $d = 5, t = 250$ and $\tau = 0.1$.

Method	$d = 5$		
	I_1	I_2	I_3
MSEM	30.51304	3.10632	9.30975
[37]	30.6888	3.12755	9.12755
[55]	30.475000	3.085333	9.563244
[59]	30.48700	3.08631	9.56594
[33]	30.4869998	3.08613	9.56533

numerical solution $w(x, t)$ for $t = 0, 5, 10, 15, 20$, with $N = 7$ and $N_e = 10$ in Fig. 6.

6.2. The undular bore

In these problems, we consider the RLWE (1.1) with initial condition

$$u(x, 0) = \frac{U_0}{2} \left(1 - \tanh \left(\frac{x - x_0}{d} \right) \right),$$

and boundary conditions

$$u(\alpha, t) = U_0, \quad u(\beta, t) = 0,$$

With the interpolation for $u(\alpha, t) = U_0$ and $u(\beta, t) = 0$ we have

$$\hat{u}(x, t) = \frac{-U_0}{\beta - \alpha} (x - \alpha) + U_0$$

In order to homogenize the boundary conditions, we consider

$$y(x, t) = u(x, t) - \hat{u}(x, t),$$

With these definitions, the undular bore, becomes as follows

$$\frac{\partial(y + \hat{u})}{\partial t} + (1 + \epsilon(y + \hat{u})) \frac{\partial(y + \hat{u})}{\partial x} - \mu \frac{\partial}{\partial t} \left(\frac{\partial^2(y + \hat{u})}{\partial x^2} \right) = 0,$$

with the new initial and boundary conditions

$$y(x, 0) = u(x, 0) - \hat{u}(x, 0) = \frac{U_0}{2} \left(1 - \tanh \left(\frac{x - x_0}{d} \right) \right) - \hat{u}(x, 0),$$

$$y(\alpha, t) = y(\beta, t) = 0.$$

We solve the undular bore problem with the following data

$$U_0 = 0.1, \quad x_0 = 0, \quad d = 2, 5, \quad \epsilon = 1.5,$$

$$\mu = \frac{1}{6} \quad 0 \leq t \leq 250, \quad -36 \leq x \leq 300.$$

In Tables 4 and 5, we give the invariant values of MSEM with those obtained in [33,37,55,59], for $N = 8, N_e = 10, \tau = 0.1, t = 250$ and $d = 2, 5$ respectively. As can be seen from these tables, MSEM results are in good agreement with other papers. We also depicted our results for $t = 150, 250$ and $d = 2, 5$ on Figs. 7 and 8, respectively.

7. Conclusion

In this paper, we used the SEM to solve the RLWE. In the SEM, we used new bases based on Müntz polynomials and proved the

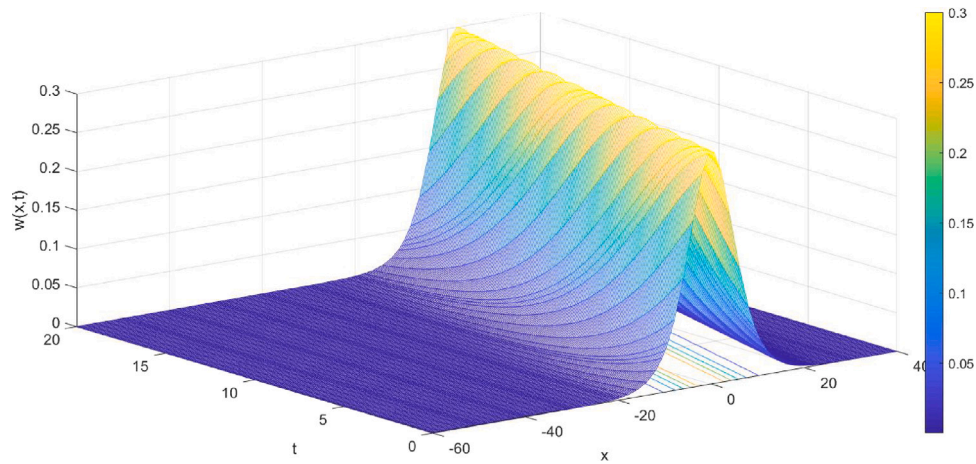


Fig. 4. MSEM approximation for (MSSW) with $N = 7$, $N_e = 10$ and $\tau = 0.1$.

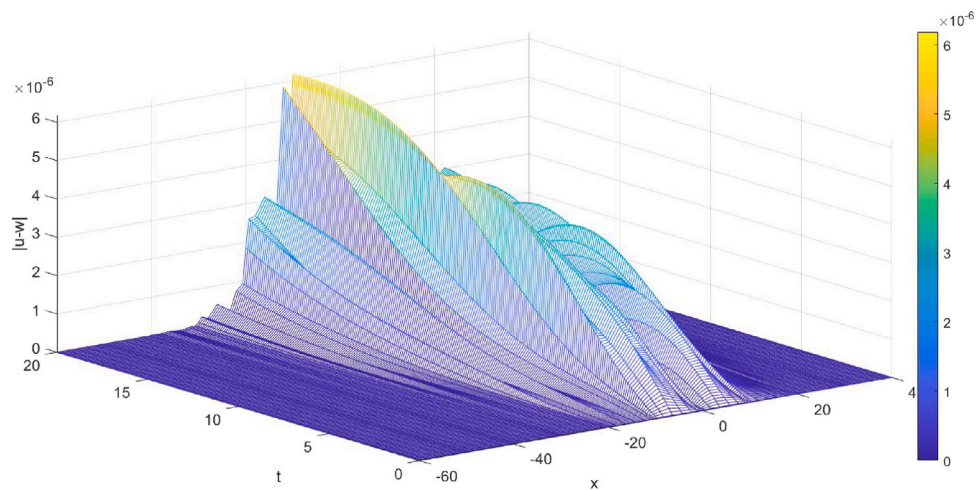


Fig. 5. Absolute error of MSEM approximation for (MSSW) with $N = 7$, $N_e = 10$ and $\tau = 0.1$.

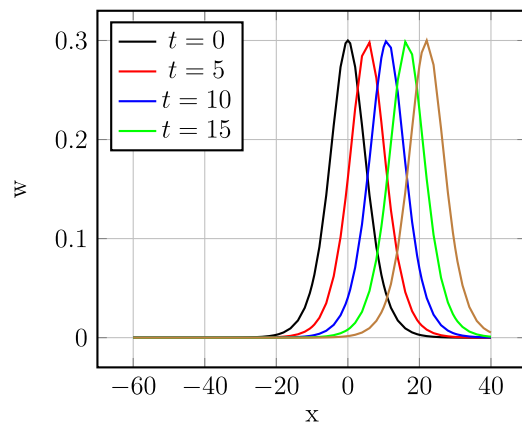


Fig. 6. MSEM approximation $w(x,t)$ for (MSSW) with $t = 0, 5, 10, 15, 20$, $N = 7$, $N_e = 10$, $c = 0.1$ and $\tau = 0.1$.

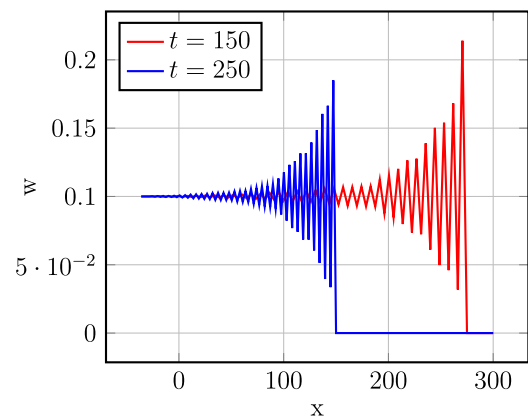


Fig. 7. MSEM approximation $w(x,t)$ for wave undulation with $N = 7$, $N_e = 10$ and $d = 2$.

convergence and stability of the method for RLWE. We compared the numerical results with previous results and showed the efficiency of the introduced method. In the MSEM, the accuracy of the numerical

solution can be increased by increasing either N or N_e . The matrices obtained from MSEM are sparse and have a smaller condition number than the LSEM method.

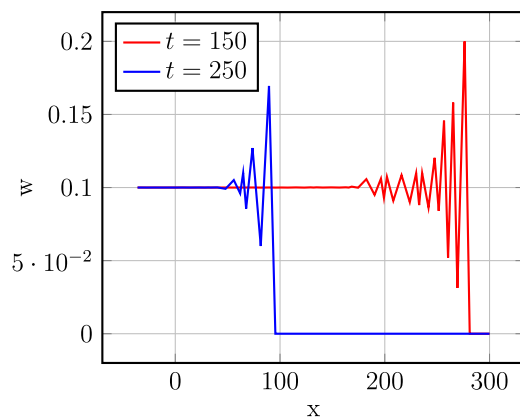


Fig. 8. MSEM approximation $w(x, t)$ for wave undulation with $N = 7$, $N_e = 10$ and $d = 5$.

CRedit authorship contribution statement

Mahmoud Lotfi: Validation, Software, Resources, Project administration, Conceptualization. **Amjad Alipanah:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization.

Ethics approval and consent to participate

The authors approve of their collaboration in writing the paper.

Consent for publication

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Not applicable.

Data availability

The datasets generated during and analyzed during the current study are available from the corresponding author on reasonable request.

References

- [1] R.D. Richtmyer, K.W. Morton, *Difference Methods for Initial-Value Problems*, second ed., Interscience Publishers, New York, 1967.
- [2] R. Eymard, T.R. Gallouët, R. Herbin, The finite volume method, in: P.G. Ciarlet, J.L. Lions (Eds.), *Handbook of Numerical Analysis*, Vol. VII, Elsevier, Amsterdam, 2000, pp. 713–1020, [http://dx.doi.org/10.1016/S1570-8659\(00\)07005-8](http://dx.doi.org/10.1016/S1570-8659(00)07005-8).
- [3] C. Hirsch, *Numerical Computation of Internal and External Flows, Volume 2: Computational Methods for Inviscid and Viscous Flows*, Wiley, Chichester, 1991, ISBN 978-0-471-92452-4.
- [4] C.B. Laney, *Computational Gas Dynamics*, Cambridge University Press, Cambridge, 1998.
- [5] R.J. LeVeque, *Numerical Methods for Conservation Laws*, second ed., Birkhäuser, Basel, 1992.
- [6] J.C. Slater, Electronic energy bands in metals, *Phys. Rev.* 45 (10) (1934) 794–801, <http://dx.doi.org/10.1103/PhysRev.45.794>.
- [7] B.G. Galerkin, On the method of solving boundary value problems in mathematical physics, *Vestnik Inzhenerov Tekhnikov* 19 (1915) 897–908.
- [8] C. Canuto, M.Y. Hussaini, A. Quarteroni, T.A. Zang, *Spectral Methods in Fluid Dynamics*, Springer-Verlag, Berlin, 1988, <http://dx.doi.org/10.1007/978-3-642-84108-8>.
- [9] A. Hrennikoff, Solution of problems of elasticity by the framework method, *J. Appl. Mech.* 8 (4) (1941) 169–175, <http://dx.doi.org/10.1115/1.4009129>.
- [10] R. Courant, Variational methods for the solution of problems of equilibrium and vibrations, *Bull. Am. Math. Soc.* 49 (1) (1943) 1–23, <http://dx.doi.org/10.1090/S0002-9904-1943-07818-4>.
- [11] G. Strang, G. Fix, *An Analysis of the Finite Element Method*, Prentice-Hall, Englewood Cliffs, NJ, 1973.
- [12] O.C. Zienkiewicz, R.L. Taylor, J.Z. Zhu, *The Finite Element Method: Its Basis and Fundamentals*, seventh ed., Butterworth-Heinemann, Oxford, 2013.
- [13] K.J. Bathe, *Finite Element Procedures*, second ed., Klaus-Jürgen Bathe, Cambridge, MA, 2014.
- [14] A.T. Patera, A spectral element method for fluid dynamics: Laminar flow in a channel expansion, *J. Comput. Phys.* 54 (3) (1984) 468–488, [http://dx.doi.org/10.1016/0021-9991\(84\)90128-1](http://dx.doi.org/10.1016/0021-9991(84)90128-1).
- [15] E. Priolo, G.A. Seriani, A numerical investigation of Chebyshev spectral element method for acoustic wave propagation, in: *Proceedings of the 13th IMACS Conference on Computational and Applied Mathematics*, IMACS, 1991, pp. 154–172.
- [16] C. Zhu, G. Qin, J. Zhang, Implicit Chebyshev spectral element method for acoustics wave equations, *Finite Elem. Anal. Des.* 47 (2) (2011) 184–194, <http://dx.doi.org/10.1016/j.finel.2010.09.004>.
- [17] Y. Li, X.K. Li, The Chebyshev spectral element approximation with exact quadratures, *J. Comput. Appl. Math.* 296 (2016) 320–333, <http://dx.doi.org/10.1016/j.cam.2015.09.021>.
- [18] A. Quaglino, M. Gallieri, J. Masci, J. Koutnik, Accelerating neural ODEs with spectral elements, 2019, arXiv preprint, [arXiv:1906.07038](https://arxiv.org/abs/1906.07038).
- [19] M. Lotfi, A. Alipanah, Legendre spectral element method for solving sine-Gordon equation, *Adv. Difference Equ.* 2019 (1) (2019) 113, <http://dx.doi.org/10.1186/s13662-019-2059-7>.
- [20] M. Saffarian, A. Mohebbi, Numerical solution of two and three dimensional time fractional damped nonlinear Klein–Gordon equation using ADI spectral element method, *Appl. Math. Comput.* 405 (2021) 126182, <http://dx.doi.org/10.1016/j.amc.2021.126182>.
- [21] M. Abbaszadeh, M. Dehghan, The proper orthogonal decomposition modal spectral element method for two-dimensional viscoelastic equation, *Thin-Walled Struct.* 161 (2021) 1–14, <http://dx.doi.org/10.1016/j.tws.2020.107429>.
- [22] M. Dehghan, N. Shafieeabyaneh, M. Abbaszadeh, Application of spectral element method for solving Sobolev equations with error estimation, *Appl. Numer. Math.* 158 (2020) 439–462, <http://dx.doi.org/10.1016/j.apnum.2020.08.010>.
- [23] T.B. Benjamin, J.L. Bona, J.J. Mahony, Model equations for long waves in nonlinear dispersive systems, *Phil. Trans. R. Soc. A* 272 (1220) (1972) 47–78, <http://dx.doi.org/10.1098/rsta.1972.0032>.
- [24] M.J. Ablowitz, P.A. Clarkson, *Solitons, Nonlinear Evolution Equations and Inverse Scattering*, Cambridge University Press, Cambridge, 1991.
- [25] D.H. Peregrine, Calculations of the development of an undular bore, *J. Fluid Mech.* 25 (2) (1966) 321–330, <http://dx.doi.org/10.1017/S0022112066001678>.
- [26] R. Mokhtari, M. Mohammadi, Numerical solution of GRLW equation using sinc-collocation method, *Comput. Phys. Comm.* 181 (7) (2010) 1266–1274, <http://dx.doi.org/10.1016/j.cpc.2010.03.015>.
- [27] Siraj-ul-Islam, S. Haq, A. Ali, A meshfree method for the numerical solution of the RLW equation, *J. Comput. Appl. Math.* 223 (2) (2009) 997–1012, <http://dx.doi.org/10.1016/j.cam.2008.03.039>.
- [28] M.N. Rasoulzadeh, M.J. Ebadi, Z. Avazzadeh, O. Nikan, An efficient local meshless method for the equal width equation in fluid mechanics, *Eng. Anal. Bound. Elem.* 131 (2021) 258–268, <http://dx.doi.org/10.1016/j.enganabound.2021.07.001>.
- [29] H.N. Hassan, K.S. Hassan, The solution of the regularized long wave equation using the Fourier leapfrog method, *Z. Naturforschung A* 65 (4) (2010) 268–276, <http://dx.doi.org/10.1515/zna-2010-0402>.
- [30] D. Bhardwaj, R.A. Shankar, A computational method for regularized long wave equation, *Comput. Math. Appl.* 40 (12) (2000) 1397–1404, [http://dx.doi.org/10.1016/S0898-1221\(00\)00248-0](http://dx.doi.org/10.1016/S0898-1221(00)00248-0).
- [31] J. Xu, S. Xie, H. Fu, A two-grid block-centered finite difference method for the nonlinear regularized long wave equation, *Appl. Numer. Math.* 171 (2022) 128–148, <http://dx.doi.org/10.1016/j.apnum.2021.08.008>.
- [32] S.E. Kutluay, A.L. Esen, A finite difference solution of the regularized long-wave equation, *Math. Probl. Eng.* 2006 (2006) 85743, <http://dx.doi.org/10.1155/MPE/2006/85743>.
- [33] A. Esen, S. Kutluay, Application of a lumped Galerkin method to the regularized long wave equation, *Appl. Math. Comput.* 174 (2) (2006) 833–845, <http://dx.doi.org/10.1016/j.amc.2005.05.032>.
- [34] D.A. Hammad, Application of Bernstein collocation method for solving the generalized regularized long wave equations, *Ain Shams Eng. J.* 12 (4) (2021) 4081–4089, <http://dx.doi.org/10.1016/j.asej.2021.04.005>.

- [35] S.K. Bhowmik, S.B.G. Karakoc, Numerical solutions of the generalized equal width wave equation using the Petrov–Galerkin method, *Appl. Anal.* 100 (4) (2021) 714–734, <http://dx.doi.org/10.1080/00036811.2019.1616696>.
- [36] L. Guo, H. Chen, L^1 -Galerkin mixed finite element method for the regularized long wave equation, *Computing* 77 (3) (2006) 205–221, <http://dx.doi.org/10.1007/s00607-005-0158-7>.
- [37] O.E. Hepson, İ. Dağ, B. Saka, B. Ay, The cubic B-spline least-squares finite-element method for the numerical solutions of regularized long-wave equation, *Int. J. Comput. Math.* 99 (2) (2022) 374–385, <http://dx.doi.org/10.1080/00207160.2021.1940979>.
- [38] C. Lu, W. Huang, J. Qiu, An adaptive moving mesh finite element solution of the regularized long wave equation, *J. Sci. Comput.* 74 (1) (2018) 122–144, <http://dx.doi.org/10.1007/s10915-017-0427-6>.
- [39] Ch. H. Müntz, in: H.A. Schwarz Festschrift (Ed.), *Über den Approximationssatz von Weierstrass*, Springer, Berlin, 1914, pp. 303–312.
- [40] O. Szász, Über die Approximation stetiger Funktionen durch lineare Aggregate von Potenzen, *Math. Ann.* 77 (1916) 482–496, <http://dx.doi.org/10.1007/BF01456964>.
- [41] G.V. Badalyan, Generalization of Legendre polynomials and some of their applications, *Izv. Akad. Nauk. Armyanskoi SSR, Seriya Fiziko-Matematicheskikh I Estestv. Nauk.* 8 (5) (1955) 1–28; 9 (1956) 3–22 (in Russian with Armenian summary).
- [42] A.K. Taslakyan, Some properties of Legendre quasipolynomials with respect to a Müntz system, *Math. Erevan Univ. Erevan* 2 (1984) 179–189 (in Russian with Armenian summary).
- [43] J. Shen, Y. Wang, Müntz–Galerkin methods and applications to mixed Dirichlet–Neumann boundary value problems, *SIAM J. Sci. Comput.* 38 (4) (2016) 2357–2381, <http://dx.doi.org/10.1137/15M1052391>.
- [44] S. Hosseinpour, A. Nazemi, E. Tohidi, Müntz–Legendre spectral collocation method for solving delay fractional optimal control problems, *J. Comput. Appl. Math.* 351 (2019) 344–363, <http://dx.doi.org/10.1016/j.cam.2018.10.058>.
- [45] S. Sabermahani, Y. Ordokhani, A new operational matrix of Müntz–Legendre polynomials and Petrov–Galerkin method for solving fractional Volterra–Fredholm integrodifferential equations, *Comput. Methods Differ. Equ.* 8 (3) (2020) 408–423, <http://dx.doi.org/10.22034/cmde.2020.32623.1515>.
- [46] P. Mokhtary, F. Ghoreishi, H.M. Srivastava, The Müntz–Legendre Tau method for fractional differential equations, *Appl. Math. Model.* 40 (2) (2016) 671–684, <http://dx.doi.org/10.1016/j.apm.2015.06.014>.
- [47] G.V. Milovanović, *Müntz Orthogonal Polynomials and their Numerical Evaluation*, in: *International Series of Numerical Mathematics*, vol. 131, Birkhäuser, Basel, 1999.
- [48] P. Borwein, T. Erdélyi, J. Zhang, Müntz systems and orthogonal Müntz–Legendre polynomials, *Trans. Amer. Math. Soc.* 342 (2) (1994) 523–542, <http://dx.doi.org/10.1090/S0002-9947-1994-1223232-7>.
- [49] A. Dogan, Application of Galerkin’s method to equal width wave equation, *Appl. Math. Comput.* 160 (1) (2005) 65–76, <http://dx.doi.org/10.1016/j.amc.2003.08.105>.
- [50] G.D. Smith, *Numerical Solution of Partial Differential Equations: Finite Difference Methods*, second ed., Clarendon Press, Oxford, 1985.
- [51] A. Quarteroni, A. Valli, Numerical Approximation of Partial Differential Equations, in: *Springer Series in Computational Mathematics*, vol. 23, Springer, Berlin, 1994, <http://dx.doi.org/10.1007/978-3-540-85268-1>.
- [52] A. Ghafoor, S. Haq, An efficient numerical scheme for the study of equal width equation, *Results Phys.* 9 (2018) 1411–1416, <http://dx.doi.org/10.1016/j.rinp.2018.04.060>.
- [53] İ. Dağ, M.N. Özer, Approximation of the RLW equation by the least square cubic B-spline finite element method, *Appl. Math. Model.* 25 (3) (2001) 221–231, [http://dx.doi.org/10.1016/S0307-904X\(00\)00030-5](http://dx.doi.org/10.1016/S0307-904X(00)00030-5).
- [54] A. Zaki, Solitary waves of the splitted RLW equation, *Comput. Phys. Comm.* 138 (1) (2001) 80–91, [http://dx.doi.org/10.1016/S0010-4655\(01\)00200-4](http://dx.doi.org/10.1016/S0010-4655(01)00200-4).
- [55] Ö. Oruç, F. Bulut, A. Esen, Numerical solutions of regularized long wave equation by haar wavelet method, *Mediterr. J. Math.* 13 (5) (2016) 3235–3253, <http://dx.doi.org/10.1007/s00009-016-0682-z>.
- [56] İ. Dağ, B. Saka, D. Irk, Galerkin method for the numerical solution of the RLW equation using quintic B-splines, *J. Comput. Appl. Math.* 190 (1–2) (2006) 532–547, <http://dx.doi.org/10.1016/j.cam.2005.04.026>.
- [57] N.M. Yagmurlu, Y. Uçar, İ. Çelikkaya, Operator splitting for numerical solutions of the RLW equation, *J. Appl. Anal. Comput.* 8 (5) (2018) 1494–1510, <http://dx.doi.org/10.11948/2018.1494>.
- [58] Ö. Oruç, A. Esen, F. Bulut, A strang splitting approach combined with Chebyshev wavelets to solve the regularized long-wave equation numerically, *Mediterr. J. Math.* 17 (4) (2020) 140, <http://dx.doi.org/10.1007/s00009-020-01572-w>.
- [59] B. Saka, İ. Dağ, A numerical solution of the RLW equation by Galerkin method using quartic B-splines, *Commun. Numer. Methods Eng.* 24 (11) (2008) 1339–1361, <http://dx.doi.org/10.1002/cnm.1036>.